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Security Operations in Public Safety

Applying the CISSP Incident Lifecycle to PSAP Cyber Resilience

This paper examines how CISSP Domain 7 – Security Operations can be applied to strengthen cyber resilience in 911 call centers. As these centers transition from legacy infrastructure to IP based next generation 911 architectures, they face growing exposure to cyber threats such as ransomware, denial of service attacks, and exploitation of compromised credentials. Using a TDoS event from 2016 as the primary case study, the paper analyzes how gaps in monitoring baselines, incident recognition, and inter-agency communication and information sharing contributed to the delayed recovery of this event. The analysis is framed around the security operations incident management lifecycle: preparation and prevention, detection and analysis, response and containment, recovery, and post incident improvement. For each phase, the paper identifies practical measures that PSAP technical leaders can implement. The paper concludes that treating cybersecurity as a core element of PSAP operations is essential to maintaining public trust and ensuring that 911 services remain available during a cyber disruption.

TABLE OF CONTENTS

Title Page	1
Table of Contents	2
Introduction	3
Cyber Threat Landscape	4
Case Study	6
Security Operations in the PSAP Environment	7
Preparation and prevention	7
Detection and Analysis	9
Response and Containment	10
Recovery and Post Incident Improvement	12
Conclusion	13
References	14

I. Introduction

Each year, approximately 240 million calls are placed to 911 call centers in the United States. These centers, also known as Public Safety Answering Points, or PSAPs, are the lifeline Americans expect to rely on over 657,000 times a day (National Emergency Number Association [NENA], n.d.). But what happens if no one answers when they call?

Today, PSAPs range in technical readiness from severely under-prepared or under-resourced centers running legacy systems to highly modernized, Next Generation 911 (NG911) environments. As voice, text, and data traffic increasingly traverse IP networks, PSAPs must integrate computer aided dispatch (CAD) systems, VoIP call handling, GIS mapping, and logging systems with broader municipal state, county, and municipal IT infrastructure. This convergence expands their attack surface and exposes mission critical services to many of the same cyber threats that target other enterprises, but with far higher stakes.

Recent years have demonstrated that these risks are more than theoretical. In 2016, an Arizona teenager shared a malicious link on social media that caused smartphones and other call capable connected devices to repeatedly dial 911 until they were powered off. Within hours, PSAPs in multiple states were flooded with calls, several centers had to shut down lines, and hundreds of legitimate callers encountered busy signals or long delays. This event, along with more recent disruptions to CAD, radio, and other PSAP systems, illustrates that cyber incidents in public safety environments can directly translate into delayed responses, responder safety risks, and preventable loss of life.

This paper applies CISSP Domain 7 – Security Operations to the PSAP context, using the Arizona telephony denial of service (TDoS) incident as a primary case study. It examines how

technical leaders in PSAPs can use the incident management lifecycle to strengthen prevention, detection, response, recovery, and continuous improvement. In doing so, it highlights specific security operations concepts such as logging and monitoring, configuration and change management, resource protection, and disaster recovery planning, that are especially critical for maintaining cyber resilience in public safety communications.

II. Cyber Threat Landscape for PSAPs

While PSAPs share many vulnerabilities with other state and local government agencies, several factors make them uniquely exposed. Many centers still rely on legacy operating systems, proprietary telephony hardware, or analog to digital gateways that were never designed with modern cyber threats in mind (SecuLore Solutions, n.d.). Because PSAPs operate around the clock, patching and system replacement often lags behind best practices. Leadership may be reluctant to schedule outages or invest in disruptive upgrades when the system appears to be “working.” At the same time, NG911 initiatives have pushed call handling, CAD, and mapping functions onto platforms that integrate with third party APIs like Pulse Point or Rapid SOS (NENA, 2020). Unfortunately, this integration often increases reliance on remote access, virtual private networks, and cloud hosted services.

Recent data suggests that cyber threat actors have become more aware of these mission critical systems, and more willing to target them. A 2025 joint report by the Multi-State Information Sharing and Analysis Center (MS-ISAC) and Motorola Solutions’ Public Safety Threat Alliance documented 326 cyberattacks against public safety organizations in 2024. Even as overall attacks against public safety entities declined by 12%, attacks impacting mission critical infrastructure

like public safety radio, CAD systems, and PSPA networks rose by 60% (Motorola PSTA & MS-ISAC, 2025). In at least 24 cases, these attacks rendered emergency communications completely unavailable, forcing agencies to revert to paper processes and backup radios for extended periods, something the author has experienced firsthand while working in a PSAP.

The general operations of PSAPs have not been spared as well. The same report identified multiple incidents where PSAP operations were degraded by ransomware, misconfigured remote access, and TDoS attacks. In one example, a TDoS against a third-party 911 call handling service caused approximately five hours of disruption across more than 20 agencies, leading to difficulty hearing callers and missing location information (Motorola PSTA & MS-ISAC, 2025). These disruptions illustrate how tightly coupled PSAPs are to vendor systems and how quickly a fault or attack in one part of the ecosystem can cascade into a larger incident.

Threat actors most commonly gain initial access through credential abuse, exploitation of unpatched vulnerabilities, and poorly secured remote services such as VPNs or exposed management interfaces (Motorola PSTA & MS-ISAC, 2025). Once inside, they often pivot from general municipal networks into mission critical environments, targeting CAD or records servers, PSAP workstations, or radio infrastructure. Other incidents, like the attack against CAD and 911 systems in Baltimore (Baltimore's 911 system hit with cyber hack, 2018) demonstrates that ransomware remains a dominant threat, but TDoS, misconfiguration abuse, and data theft operations also pose serious risks to PSAP availability and confidentiality.

For PSAP technical leaders, this threat landscape reinforces that cybersecurity is not purely an IT concern. Any cyber event that affects 911 call handling, dispatch, or responder communications becomes a life safety incident. As such, security operations practices must be adapted to the PSAP environment and treated as operational necessities rather than optional improvements.

III. Case Study: The 2016 TDoS Attack

The 2016 Arizona TDoS incident provides a useful case study for understanding how cyberattacks can rapidly overwhelm PSAP operations. In this event, a teenager created and shared a short URL on Twitter that, when clicked on by a smartphone or other capable device, triggered repeated 911 calls without the user's consent or control. The only way to stop the device from dialing 911 was to power off the device (Fletcher, 2016). The link spread widely across social media, and within hours, PSAPs in Arizona and at least six other states experienced massive surges in call volume.

From an operational perspective, the symptoms initially resembled ordinary trunk congestion or a misbehaving routing configuration. Call volume spiked, lines became saturated, and some centers reported that legitimate callers were unable to reach an operator. Because no single PSAP had immediate visibility into the broader pattern, some agencies believed they were experiencing isolated technical issues rather than a coordinated attack. As a result, early response actions focused on troubleshooting local equipment and engaging telephony vendors one on one, instead of rapidly classifying the situation as a cyber incident.

Once state-level officials and carriers recognized the common pattern across multiple centers, they coordinated with PSAPs to trace back to the malicious link, adjust call handling, and mitigate the impact. Some centers temporarily shut down affected trunks or rerouted traffic to neighboring PSAPs. Others implemented manual triage, prioritizing calls that appeared to be from local numbers while leaving suspected automated calls in queue. Even with these measures,

the incident resulted in hundreds of missed or delayed calls and highlighted significant gaps in monitoring, cross agency communication, and incident response (Fletcher, 2016).

Viewed through the lens of CISSP's Domain 7, the Arizona TDoS event illustrates several key lessons. First, PSAPs lacked pre-defined baselines for what "abnormal" call behavior looked like across multiple centers, making it harder to distinguish malicious traffic from routine variation. Second, incident escalation paths between local PSAPs, state 911 authorities, and carriers were not clearly documented or rehearsed, leading to delays in declaring a coordinated incident. Finally, there was limited guidance on how to balance call blocking, rerouting, and overflow strategies while safeguarding access for legitimate callers.

These gaps translate directly to security operations concepts like preparation and prevention, detection and analysis, response and containment, recovery, and lessons learned. Applying the domain 7 incident management lifecycle to this case offers a framework for how PSAP leaders can strengthen their cyber resilience against future TDoS and related threats.

IV. Applying CISSP Security Operations in the PSAP Environment

Preparation and Prevention

Preparation and prevention form the foundation of effective security operations in any environment, but doubly so in a 911 center environment. PSAPs cannot afford extended downtime for maintenance or troubleshooting, so they must identify, mitigate, and document cyber risks before an incident occurs.

A critical first step is establishing operational baselines and monitoring practices. PSAPs should define normal ranges for call volume, trunk utilization, CAD performance, and network traffic, then configure logging and monitoring systems such as security information and event management (SIEM) tools or call volume analytics to alert when those metrics deviate significantly from expected patterns (NIST, 2018). These logs also become important forensic artifacts during and after an incident, allowing investigators to reconstruct events and preserve evidence for potential legal or regulatory review (Nelson, 2025).

Preparation also includes staff training and exercises. Security operations emphasize that incident management is as much a human process as a technical one. Regular cybersecurity awareness training helps dispatchers, supervisors, and IT staff recognize phishing, social engineering, and suspicious system behavior. Tabletop and simulation-based exercises built around realistic events like TDoS or ransomware scenarios allow PSAP personnel to rehearse incident identification and classification, escalation procedures, and interim workflows such as manual call handling or backup radio protocols. These exercises double as disaster recovery plan tests, validating that documented procedures are practical and effective (Nelson, 2025).

Configuration and change management are equally important. PSAP environments often include a mix of integrated servers and workstations running various resources. Maintaining an accurate asset inventory, standard configurations, and documented change histories makes it easier to identify anomalies, roll back failed changes, and understand which systems may be affected by vulnerability or misconfiguration. Security operations concepts such as least privilege, separation of duties, and privileged account management apply directly here. Access to PSAP systems should be restricted, logged, and periodically reviewed (Nelson, 2025); no single individual should be able to introduce unapproved changes without oversight.

Patch and vulnerability management must be tailored to the PSAPs limited maintenance windows. Rather than applying updates ad hoc, technical leaders can define risk-based schedules that prioritize cycles during low call volume windows and target the most vulnerable systems first. Coordinating patching with vendors and including clear patch expectations in service level agreements (SLAs) can help reduce lag in applying patches without compromising availability (Nelson, 2025).

Finally, preparation requires formal documentation. An incident response plan (IRP), continuity of operations plan (COOP), and disaster recovery plan (DRP) should clearly define roles and responsibilities, decision thresholds for declaring an incident, communication paths with state authorities and carriers, and criteria for activating alternate call routing or backup sites (Swanson et al., 2010). These plans should also address evidence collection, chain of custody for logs and recordings, and reporting obligations to stakeholders or oversight bodies.

Detection and Analysis

Once an incident begins, the speed and accuracy with which a PSAP can recognize abnormal conditions largely determines how serious the impact will be. In the Arizona TDoS event, early symptoms such as trunk congestion and unusual call patterns were initially misinterpreted. A more mature security operations posture would treat these anomalies as potential indicators of compromise and trigger closer analysis.

Detection in the PSAP context depends on both automated tools and frontline observations. SIEM platforms, intrusion detection systems (IDS), and specialized telephony monitoring tools can generate alerts when performance deviates from those expected baselines, or centers start

receiving calls from unusual sources. At the same time, dispatchers and supervisors may notice that calls are silent, that caller ID information appears suspicious, or that location data is missing entirely. It is important to capture and recognize these technical and human signals as they can inform the incident management process.

Once alerts are generated, analysts must quickly triage them. In a PSAP, triage often means distinguishing between a call spike due to a real-world event like a major crash or highly visible fire, and an artificial spike due to malicious traffic. Predefined criteria for declaring a cyber incident can reduce hesitation and promote consistent decision making. Threat intelligence services and sector specific information sharing organizations can also assist by confirming whether similar anomalies are being observed by other agencies (NENA, 2020).

Effective analysis relies on quality data. Logging configurations should ensure that call records, session-initiated protocol (SIP) headers, firewall logs, VPN connections, and system events are retained with sufficient detail and timestamps to support investigation. When an incident is suspected evidence handling practices such as preserving copies of critical logs, documenting who accessed systems, and avoiding destructive troubleshooting steps will help to maintain the integrity of forensic artifacts while still helping containment and recovery efforts.

Response and Containment

Once a PSAP confirms it is experiencing a cyber incident, response priorities shift to preserving 911 service while preventing the attacker from expanding their impact. For security staff, this often means balancing the desire to aggressively block malicious traffic against the risk of inadvertently preventing legitimate calls from reaching an operator.

Well-designed COOP and IRP documents, rehearsed tabletop and simulation exercises give PSAP leadership a playbook of pre-approved actions. In a TDoS event, these may include temporarily rerouting some trunks to partner PSAPs, enabling overflow lines, or manually prioritizing calls based on certain caller ID or location indicators. In a ransomware or network compromise scenario, response actions might focus on quarantining affected systems, disabling certain remote access paths, or switching to manual dispatch and backup procedures while affected resources are remediated in isolation.

Technical containment measures rely heavily on maintaining detective and preventative controls. Firewalls and session border controls can be configured to selectively block or rate limit traffic that matches known malicious patterns, while IDS/IPS signatures may be uploaded to recognize specific TDoS behaviors. Remote access services should be closely scrutinized; if an incident involves credential abuse or VPN exploitation, temporarily disabling non-essential remote pathways while enforcing multi-factor authentication for privileged accounts can help prevent further lateral movement.

Effective response requires coordination beyond the PSAP itself. Incident commanders must quickly notify city or county IT leadership, state 911 authorities, law enforcement partners, and telecommunication carriers. For larger events, public safety entities may also work with state or federal cyber incident response teams. Security Operations require clear reporting and documentation. Incident logs should record when anomalies were detected, what decisions were made and by whom, and how systems were reconfigured or isolated. This documentation will later support after action reviews (AAR), compliance reporting, and any necessary legal proceedings.

Throughout response and containment, personnel safety and service continuity remain paramount. Responders in the field depend on timely, accurate information to do their job. Sudden shifts to manual backup procedures can introduce new risks if staff have not been trained and given clear guidance. Including human factors and safety considerations in the incident response plan, rather than treating the response as purely technical, helps anticipate communication issues and reduce errors during a high stress incident.

Recovery and Post Incident Improvement

After an incident has been contained, PSAPs must safely restore normal operations while learning as much as possible from the event. Recovery is more than turning systems back on; it involves validating system integrity, confirming that vulnerabilities have been addressed, and ensuring that data and media remain trustworthy.

Technical recovery steps often include gradually re-enabling disabled trunks, restoring standard call routing, bringing CAD and records servers and workstations back online, and reconnecting PSAP systems to broader municipal networks. Before reintroducing traffic, security operations best practices call for checking system logs, configuration baselines, and anti-malware reports to ensure that no indicators of compromise remain. Backups may be restored when necessary, and integrity checks of critical databases, records, and mapping data should be completed.

At the same time, PSAPs must fulfil documentation and reporting requirements. Incident reports to stakeholders, oversight boards, or state and federal partners may be required, along with notifications to vendors and mutual aid partners. Chain of custody records for log files, system forensic images, and other artifacts should be maintained if legal action is possible.

Post incident improvement begins with the AAR that includes technical staff, supervisors, and in larger events possibly even representatives from vendors, carriers, and other partner agencies.

The goal is to answer not just “what happened” but also “why it happened” and “how we can prevent or better manage it in the future.” Questions might focus on whether detection thresholds were appropriate, whether escalation paths functioned as intended, if communication between agencies was timely and effective, and how well manual fallback procedures worked if required.

Findings from the AAR should feed directly into updates of the IRP, COOP, DRP, configuration standards, and training program. For example, if a TDoS incident revealed that PSAPs lacked shared visibility into statewide call patterns, in improvement might involve deploying a shared dashboard or implementing specific threat intelligence and information tracking processes that would improve that visibility across multiple centers. If privileged accounts were found to be overly broad or inadequately monitored, privileged access management policies might be adjusted to improve regulation and mitigate that risk moving forward.

Domain 7 also encourages organizations to treat each incident as an opportunity to raise the organization’s overall cyber maturity. For PSAPs, this might entail moving from informal, undocumented procedures towards standardized, tested, and periodically audited security operations practices. Incorporating lessons learned into ongoing training, new-hire onboarding, and recurring disaster recovery exercises ensure that institutional knowledge is not lost when staff rotate or retire.

V. Conclusion

PSAPS operate in a unique intersection of legacy infrastructure, cutting edge technology, public trust, and life-safety. As cyber threats targeting public safety systems continue to mature, applying the CISSP security operations concepts to PSAP environments is no longer optional. The Arizona TDoS incident and more recent disruptions demonstrate how quickly a cyber event can degrade emergency communications and how difficult it can be to recover when monitoring, coordination, and incident management processes are under developed.

By adopting a security operations lifecycle; preparing and preventing, detecting and analyzing, responding and containing, and then recovering and improving, PSAP leaders can systematically strengthen their defenses. Practical steps include establishing monitoring baselines and log management processes, enforcing least privilege and sound configuration management, rehearsing incident and disaster recovery plans through tabletop and simulation exercises, tightening patch and vulnerability management around mission critical systems, and documenting incidents in ways that support both accountability and learning.

Ultimately, cybersecurity in the PSAP environment is inseparable from public safety. Every missed or delayed 911 call during a cyber incident represents a potential life not saved or crime not stopped. PSAPs that embrace security operations principles and invest in continuous improvement will be better positioned to withstand disruption, coordinate with partners, and ensure that when any of us dials 911, help will be within reach.

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